

Notes: Bolton & Scharfstein (JPE, 1996)

Optimal Debt Structure and the Number of Creditors

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1 Big picture

- **Question.** What determines the optimal structure of corporate debt, especially the number of creditors, the allocation of collateral or security interests, and the voting rules that govern renegotiation after default?
- **Core insight.** Debt structure affects the bargaining game after default. That ex-post bargaining outcome feeds back into the manager's ex-ante incentive to default strategically.
- **Two types of default.** A **strategic default** occurs when the manager has cash but tries to divert it. A **liquidity default** occurs when the firm genuinely lacks cash.
- **Why debt is useful.** Giving creditors liquidation rights deters strategic default. Without penalties after nonpayment, a manager would always claim no cash and divert resources.
- **Why debt is costly.** Liquidation rights are inefficient when the firm suffers a true liquidity default. The manager is the highest-value user of the assets, so liquidation destroys surplus.
- **Why multiple creditors can be useful.** When assets are complementary, multiple secured creditors create holdup. The manager must pay more to buy back assets after strategic default, lowering his payoff from default.
- **Why multiple creditors can be costly.** The same holdup lowers expected liquidation value after a liquidity default because outside buyers may not incur the cost of becoming informed if they expect to be held up by several creditors.
- **Optimal trade-off.** Good debt structure intentionally balances two goals: make strategic default unattractive while keeping unavoidable liquidity default from being too inefficient.
- **Main number-of-creditors result.** Low-default-risk firms are more likely to use multiple creditors; high-default-risk firms are more likely to use one creditor to maximize liquidation value.
- **Asset complementarity result.** The formal proposition states that two creditors are optimal when complementarity is low and one creditor when complementarity is high. Small complementarity creates useful discipline, but large complementarity makes liquidity-default sales too costly.
- **Redeployability/liquidity result.** When outside buyers value assets highly, or have liquidity in bad states, one creditor is attractive because high liquidation values matter. When outside buyers have low valuations or little liquidity, multiple creditors are more attractive because the sacrifice in liquidation value is smaller.
- **Security allocation.** Giving one creditor all security replicates the one-creditor solution; spreading security interests across creditors replicates the multiple-creditor solution.
- **Voting rules.** More stringent voting rules, especially supermajority rules, make renegotiation harder and reduce the manager's strategic-default payoff. They are optimal when default risk is low or outside buyer values are low.
- **Economic takeaway.** Inefficient renegotiation is not always bad. The optimal debt contract may deliberately make renegotiation difficult because difficulty disciplines managers before default.

2 Model primitives and measurement (assets, defaults, and renegotiation)

2.1 Two assets and two periods

- At date 0, the project requires an investment cost K to buy two physical assets, A and B .
- The manager has no wealth and must raise external financing.
- At date 1, the project generates cash flow x with probability θ and zero with probability $1 - \theta$.
- At date 2, if the original manager continues, the assets generate y .
- If an outside manager uses the assets, they generate αy , with $0 < \alpha \leq 1$.
- Investors themselves generate zero if they operate the assets; they can only obtain value by selling to someone else.

Interpretation. The original manager is the highest-value user of the assets. Liquidation is therefore a punishment device, not an efficient operating choice.

2.2 Incomplete contracts

- Cash flow is observable to the manager and investors but not verifiable to courts.
- Therefore, contracts cannot be written directly on the realized cash flow.
- Physical assets are harder to divert and can be contracted on.
- Debt contracts can give creditors liquidation rights after nonpayment.

Interpretation. The model is built on the Hart-Moore distinction between cash flow and assets. Cash can be hidden or diverted; assets can be seized.

2.3 Liquidity default versus strategic default

- **Liquidity default:** cash flow is zero, so the manager cannot pay even if he wants to.
- **Strategic default:** cash flow is x , but the manager pretends it is zero and diverts the cash.
- The contract must punish nonpayment enough to deter strategic default.
- The same punishment is wasteful when nonpayment reflects a true liquidity default.

Interpretation. The optimal contract uses liquidation as an incentive device. The tension is that the signal “no payment” is noisy: it may mean no cash, or it may mean cash diversion.

2.4 Probabilistic liquidation

- The optimal contract gives creditors the right to liquidate with probability β_0 after zero payment.
- It is never optimal to liquidate after the manager makes the high-state payment R_x .
- It is also optimal to set the low-state payment $R_0 = 0$.
- Partial liquidation is not optimal in the appendix: because assets are complementary, it is better to liquidate all assets with lower probability than one asset with higher probability.

Interpretation. Probabilistic liquidation is a substitute for a harsh standard debt contract. It supplies enough discipline while minimizing the expected loss from inefficient liquidation.

2.5 Renegotiation after default

- If creditors obtain liquidation rights, they may bargain with the manager or with an outside buyer.
- The manager’s payoff after strategic default is denoted $S(n)$, where n is the number of creditors.
- The creditors’ expected liquidation payoff after a liquidity default is $L_0(n)$.
- Debt structure matters because it changes both $S(n)$ and $L_0(n)$.

Interpretation. A good debt structure lowers $S(n)$, making strategic default less attractive, but keeps $L_0(n)$ high, making liquidity default less costly.

2.6 Asset complementarity

Let y^A be the date-2 cash flow from asset A alone, and y^B the date-2 cash flow from asset B alone. The paper defines asset complementarity as

$$\Delta = y - y^A - y^B > 0. \quad (1)$$

Interpretation. If $\Delta > 0$, the assets are worth more together than apart. This creates holdup power for each creditor when each is secured by a different asset.

3 Models and theoretical specifications

3.1 Manager and investor payoffs

The manager's expected payoff under a general contract is

$$\theta[x - R_x + (1 - \beta_x)y] + (1 - \theta)[-R_0 + (1 - \beta_0)y]. \quad (2)$$

The investors' expected payoff is

$$\theta(R_x + \beta_x L_x) + (1 - \theta)(R_0 + \beta_0 L_0) - K. \quad (3)$$

The high-cash-flow incentive constraint is

$$x - R_x + (1 - \beta_x)y \geq x + \beta_0 S + (1 - \beta_0)y. \quad (4)$$

Interpretation. The manager must prefer paying R_x in the good state to pretending the cash flow is zero and renegotiating after default.

3.2 Optimal liquidation probability

The paper shows that $\beta_x = 0$ and $R_0 = 0$. The incentive constraint becomes

$$R_x \leq \beta_0(y - S). \quad (5)$$

The investors' break-even condition implies

$$\beta_0\{\theta(y - S) + (1 - \theta)L_0\} - K = 0. \quad (6)$$

Thus the optimal liquidation probability is

$$\beta_0 = \frac{K}{\theta(y - S) + (1 - \theta)L_0}. \quad (7)$$

Interpretation. The denominator is the maximum expected gross payoff creditors can get from the contract. A lower manager renegotiation rent S and a higher liquidation value L_0 both reduce the required liquidation probability.

3.3 Expected surplus loss

Using the binding constraints, the manager's expected payoff is

$$\theta x + y - K - (1 - \theta)\beta_0(y - L_0). \quad (8)$$

The efficiency loss is therefore

$$(1 - \theta)\beta_0(y - L_0). \quad (9)$$

Interpretation. The first-best surplus would avoid liquidation. The agency cost is the probability of liquidity default times the probability of liquidation times the value lost by not keeping the manager in control.

3.4 One creditor

With one creditor, an outside buyer who incurs bargaining cost c and values the whole asset bundle at αy bargains when $c \leq \frac{1}{2}\alpha y$. With c uniformly distributed on $[0, \bar{c}]$, the expected liquidation payoff is

$$L_0(1) = \frac{\alpha^2 y^2}{4\bar{c}}. \quad (10)$$

After a strategic default, the original manager bargains with the single creditor and splits surplus:

$$S(1) = \frac{y}{2}. \quad (11)$$

The one-creditor maximum expected gross creditor profit is

$$\pi(1) = \theta \frac{y}{2} + (1 - \theta) \left(\frac{\alpha^2 y^2}{4\bar{c}} \right), \quad \beta_0(1) = \frac{K}{\pi(1)}. \quad (12)$$

Interpretation. One creditor gives high liquidation value because a buyer can acquire the whole asset bundle in one negotiation. But one creditor also gives the manager a relatively high renegotiation rent after strategic default.

3.5 Two creditors with separate security interests

Suppose creditor a is secured by asset A and creditor b by asset B . With asset complementarity, the two creditors together can extract more from a buyer during bargaining. The expected liquidation payoff becomes

$$L_0(2) = \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}} < L_0(1). \quad (13)$$

The manager's payoff after strategic default is

$$S(2) = \frac{y}{2} - \frac{\Delta}{6} < S(1). \quad (14)$$

The two-creditor maximum gross payoff is

$$\pi(2) = \pi(1) + \theta \frac{\Delta}{6} - (1 - \theta) \frac{\alpha^2 \Delta^2}{36\bar{c}}. \quad (15)$$

Interpretation. Two creditors lower the manager's strategic-default payoff, but also lower expected liquidation value. This is the central debt-structure trade-off.

3.6 Propositions on the number of creditors

- **Default risk.** The firm borrows from two creditors when default risk is low (θ high) and from one creditor when default risk is high (θ low).
- **Asset complementarity.** The formal proposition states that the firm borrows from two creditors when complementarity is low and from one creditor when complementarity is high.
- **Outside buyer valuation.** The firm borrows from two creditors when outside buyers place low valuations on assets (α small) and from one creditor when outside buyer valuations are high (α high).
- **Cyclicity/liquidity extension.** If outside buyers lack cash in bad states, expected liquidation values are low anyway, so the cost of multiple creditors is smaller. This makes multiple creditors more attractive in cyclical or common-shock industries.

Interpretation. Multiple creditors are most valuable when strategic-default discipline matters more than maximizing liquidation value.

3.7 Security allocation

If only creditor a is secured, the model has one creditor with whom the manager or outside buyer must negotiate. The relevant constraints are

$$\theta R_a^x + (1 - \theta) \beta_0 L_0(1) = K^A, \quad (16)$$

$$\theta R_b^x = K^B, \quad (17)$$

$$R_a^x + R_b^x = \beta_0 [y - S(1)]. \quad (18)$$

Combining these gives

$$\beta_0 \{ \theta [y - S(1)] + (1 - \theta) L_0(1) \} = K. \quad (19)$$

Interpretation. Giving one creditor all security replicates the one-creditor outcome. Giving both creditors security replicates the two-creditor outcome. Collateral design is therefore another way to tune renegotiation difficulty.

3.8 Voting rules

Let there be n creditors and suppose that m creditors must agree to a sale. Define

$$v = \frac{m}{n+1}. \quad (20)$$

The manager's payoff after strategic default is

$$S(m, n) = (1 - v)y. \quad (21)$$

The creditors' expected liquidation payoff is

$$L_0(m, n) = v(1 - v) \frac{\alpha^2 y^2}{\bar{c}}. \quad (22)$$

The expected efficiency loss is

$$EL(m, n) = (1 - \theta)K \frac{1 - v(1 - v)(\alpha^2 y / \bar{c})}{v\{\theta + (1 - \theta)v(1 - v)(\alpha^2 y / \bar{c})\}}. \quad (23)$$

The first-order condition for the optimal v is

$$\theta v^2 \frac{\alpha^2 y}{\bar{c}} - \theta - (1 - \theta)(1 - 2v) \frac{\alpha^2 y}{\bar{c}} = 0. \quad (24)$$

Interpretation. Majority rule maximizes liquidation value in the uniform-cost case, but the optimal rule is more stringent because a higher threshold lowers the manager's renegotiation rent. Thus the optimal rule is more stringent than the rule that maximizes liquidation value.

4 Conceptual identification and theoretical design

4.1 What identifies the mechanism

- The model separates strategic default from liquidity default.
- Strategic-default discipline comes through reducing the manager's continuation payoff after renegotiation.
- Liquidity-default efficiency comes through maximizing liquidation value when the firm genuinely cannot pay.
- Debt structure matters only because it changes ex-post renegotiation payoffs.

Interpretation. The paper's logic is comparative institutional analysis: change the number of creditors, collateral allocation, or voting rule, then compare how renegotiation changes incentives and liquidation values.

4.2 Why multiple creditors discipline managers

- With complementary assets, a buyer's valuation of one asset rises after he controls the other.
- Each secured creditor can capture part of the complementarity surplus.
- In a strategic default, the original manager must buy back the assets.
- More creditors therefore lower the manager's payoff from default and allow a larger high-state payment.

Interpretation. Multiple creditors create a holdup problem. The paper's key insight is that this holdup problem is useful when it deters strategic default.

4.3 Why multiple creditors can reduce liquidation value

- In a liquidity default, the outside buyer must decide whether to incur the cost of becoming informed and bargaining.
- If many creditors can extract much of the surplus, the buyer's expected payoff from entering the bargain falls.
- Therefore, the buyer may not enter, and expected liquidation value falls.

Interpretation. The same holdup that disciplines managers can destroy value in true financial distress.

4.4 Why creditor number depends on credit quality

- If θ is high, liquidity default is unlikely. The firm can afford a structure that makes liquidation less efficient because it rarely expects to use liquidation.
- If θ is low, liquidity default is likely. Maximizing liquidation value is more important, so one creditor is better.

Interpretation. This gives the credit-quality prediction: high-quality firms can tolerate and benefit from harder renegotiation; low-quality firms should keep renegotiation and liquidation simpler.

4.5 Why asset complementarity matters

- Higher Δ lowers the manager's strategic-default rent under two creditors.
- Higher Δ also lowers expected liquidation value under two creditors.
- The formal result is threshold-like: some complementarity is useful for discipline, but high complementarity makes liquidity-default liquidation too costly.

Interpretation. Complementarity is a double-edged sword. It creates bargaining power against the manager, but also creates bargaining power against efficient outside buyers.

4.6 Why voting rules matter

- A lenient rule helps creditors complete asset sales and raises liquidation value.
- A stringent rule lowers the manager's payoff from renegotiation.
- The optimal rule is more stringent than the liquidation-value-maximizing rule.
- Stringency rises when default risk is low and when outside buyer valuations are low.

Interpretation. Voting covenants are debt-governance instruments. They are not mechanical legal details; they are chosen to shape ex-post bargaining.

4.7 What the theory cannot fully prove

- The model has only two periods and two assets in the main analysis.
- Cash-flow unverifiability, Nash/Shapley bargaining, and uniform bargaining costs are simplifying assumptions.
- The model abstracts from taxes, risk aversion, information asymmetry about asset quality, and rich bankruptcy law.
- The link between one creditor and bank debt, or many creditors and public debt, is suggestive rather than literal.
- The paper offers comparative statics, not direct empirical tests.

Interpretation. The model's contribution is a clean mechanism: ex-post renegotiation inefficiency can be an ex-ante disciplinary device.

5 Tables and figures

The paper contains no standalone empirical tables or figures. To follow the same note format and preserve the original visuals, this section uses compact **directly cropped equation panels** from the paper, grouped side by side.

5.1 Original equation panels 1–2: core contract and optimal liquidation

Read:

- Panel 1 states the manager payoff, investor payoff, and high-state incentive constraint.

- Panel 2 shows the reduced incentive constraint, binding break-even condition, manager payoff, and optimal liquidation probability.
- The key object is $\beta_0 = K/[\theta(y - S) + (1 - \theta)L_0]$.

Economic message. The contract solves an incentive problem by threatening liquidation after zero payment. Debt structure matters because it changes S and L_0 .

Given this contract, the firm's expected payoffs are

$$\theta[x - R_x + (1 - \beta_x)y] + (1 - \theta)[-R_0 + (1 - \beta_0)y]. \quad (1)$$

The investor's expected profits are

$$\theta(R_x + \beta_x L_x) + (1 - \theta)(R_0 + \beta_0 L_0) - K, \quad (2)$$

where L_x and L_0 refer to the liquidation value of the assets in the hands of the creditor when cash flow is x and zero, respectively. Although investors get no direct value from owning the assets, we shall see later that they can sell the assets. For the moment, we write the proceeds of this asset sale as L_x and L_0 .

Given that the manager has no wealth of his own, payments cannot exceed the available funds at date 1, that is, $R_0 \leq 0$ and $R_x \leq x$. These payments must also satisfy an incentive constraint that ensures that the manager has an incentive to pay R_x when cash flow is x . One can write this incentive constraint as

$$x - R_x + (1 - \beta_x)y \geq x + \beta_0 S + (1 - \beta_0)y. \quad (3)$$

Two points are worth noting about this constraint. First, S denotes the utility the manager receives from paying out R_0 when cash flow

Core payoff and incentive equations

Given these results, the incentive constraint reduces to

$$R_x \leq \beta_0(y - S). \quad (4)$$

This constraint must be binding at an optimum; otherwise it would be optimal to set $\beta_0 = 0$ and the manager would have no incentive to pay out any cash flow. The nonnegative profit constraint must also be binding; otherwise R_x could be reduced while making the manager better off and still satisfying the constraint. Thus substituting the incentive constraint into the nonnegative profit constraint implies that

$$\beta_0[\theta(y - S) + (1 - \theta)L_0] - K = 0. \quad (5)$$

Using (5), we can write the manager's expected payoff as

$$\theta x + y - K - (1 - \theta)\beta_0(y - L_0). \quad (6)$$

The first three terms in (6) sum to the net present value of the project in the first-best case of no liquidation. The last term is the expected efficiency loss from contractual incompleteness; it is the probability of a liquidation, $(1 - \theta)\beta_0$, times the loss in value from a liquidation, $y - L_0$. Given that the manager's expected payoff is decreasing in β_0 and the creditor's profits are increasing in β_0 , the optimal solution is to set β_0 equal to its minimum feasible level, that is, the solution to (5). Thus

$$\beta_0 = \frac{K}{\theta(y - S) + (1 - \theta)L_0}. \quad (7)$$

Optimal liquidation probability

5.2 Original equation panels 3–4: one-creditor solution

Read:

- With one creditor, outside buyers face a single bargaining counterparty.
- The expected liquidity-default payoff is $L_0(1) = \alpha^2 y^2 / (4\bar{c})$.
- The strategic-default renegotiation rent is $S(1) = y/2$.
- The one-creditor expected gross payoff is $\pi(1) = \theta y/2 + (1 - \theta)L_0(1)$.

Economic message. One creditor is good for efficient liquidation but weaker for strategic-default discipline.

Now suppose that cash flow is zero and the firm defaults. Control of the assets passes to the creditor. We assume Nash bargaining so that the buyer and the creditor share the surplus αy equally provided that the buyer incurs the transaction cost. Thus the buyer gets $\frac{1}{2}\alpha y$ and the creditor receives the remaining $\frac{1}{2}\alpha y$. It pays for the buyer to bargain provided that $c \leq \frac{1}{2}\alpha y$; the assets are therefore sold with probability $(\frac{1}{2}\alpha y)/\bar{c}$ assuming, as we do, that $\frac{1}{2}\alpha y < \bar{c}$. So, at date 0, the expected value of the liquidation payoff when cash flow is zero is given by

$$L_0(1) = \frac{\alpha^2 y^2}{4\bar{c}}. \quad (8)$$

One should keep in mind that we have assumed that the buyer has the up-front cash to pay for the asset at date 1. Given that cash flow

One-creditor liquidation payoff

the alternative assumption that the outside buyer does bargain. Nothing of substance changes.

Without the outside buyer, there are two parties to the bargain, so the manager and the creditor split the surplus of y equally according to Nash bargaining. Thus

$$S(1) = \frac{y}{2}. \quad (9)$$

Note that the manager does not incur a cost to bargain. The assumption that the cost is zero is just a normalization and reflects the idea that the manager is more informed about the firm and therefore has a lower cost of preparing a bid.

We can now use (8) and (9) to write β_0 in terms of the exogenous parameters, α , y , and $\bar{c}$. Specifically, $\beta_0(1) = K/\pi(1)$, where

$$\pi(1) \equiv \theta \frac{y}{2} + (1 - \theta) \left(\frac{\alpha^2 y^2}{4\bar{c}} \right). \quad (10)$$

It is straightforward to establish that $\beta_0(1)$ is decreasing in θ so

Manager rent and creditor gross payoff

5.3 Original equation panels 5–6: two-creditor solution

Read:

- With two secured creditors, complementarity gives each creditor holdup power.
- The manager's strategic-default rent falls to $S(2) = y/2 - \Delta/6$.
- The liquidity-default payoff falls to $L_0(2) = L_0(1) - \alpha^2\Delta^2/(36\bar{c})$.
- The net gain from two creditors depends on whether the discipline term $\theta\Delta/6$ beats the liquidation-cost term $(1 - \theta)\alpha^2\Delta^2/(36\bar{c})$.

Economic message. Two creditors are useful precisely because they can be difficult to coordinate. That difficulty disciplines strategic default but can harm true liquidation.

Example in management bargaining. These effects are motivating, for example, by Gertner (1990) in an extensive-form bargaining game with three players, asymmetric information, and asset complementarity.

It follows then that the outside buyer's Shapley value is $\frac{1}{2}\alpha y - \frac{1}{6}\alpha\Delta$ in the event he bargains, and therefore he will bargain with probability $(1/\bar{c})(\frac{1}{2}\alpha y - \frac{1}{6}\alpha\Delta)$. So the expected payoff to the two creditors is

$$\begin{aligned} L_0(2) &= \frac{1}{\bar{c}} \left(\frac{\alpha y}{2} + \frac{\alpha\Delta}{6} \right) \left(\frac{\alpha y}{2} - \frac{\alpha\Delta}{6} \right) \\ &= \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}} \\ &< \frac{\alpha^2 y^2}{4\bar{c}} \\ &= L_0(1). \end{aligned} \tag{11}$$

Thus two creditors receive more than one if the outside buyer enters the bargain, but they receive less from an ex ante perspective since they are less likely to find a buyer of the assets. The reason is that the low payoff of making the bid discourages the outside buyer from incurring the cost c of bidding. That the latter effect is strong enough to outweigh the former follows from our assumption that the transaction costs are uniformly distributed. We discuss the importance of these assumptions later in this section.

The calculation of $S(2)$ follows similarly. Again, there are three parties to the bargain: the outside buyer and the two creditors. Since the manager is the most efficient user of the assets, he will end up buying them back from the creditors. The manager's Shapley value is given by

$$S(2) = \frac{y}{2} - \frac{\Delta}{6}. \tag{12}$$

Two-creditor liquidation payoff and manager rent

Shapley value when there is only one creditor, $\frac{1}{2}y$. As in the case of a liquidity default, asset complementarity forces the buyer (in this case the manager) to pay more because control of an asset is worth more if the buyer already owns the complementary asset.²

Given (7), we know that $\beta_0(2) = K/\pi(2)$, where

$$\pi(2) = \theta \left(\frac{y}{2} + \frac{\Delta}{6} \right) + (1 - \theta) \left(\frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}} \right) \tag{13}$$

Two-creditor gross payoff

5.4 Original equation panels 7–8: formal propositions on creditor number

Read:

- Proposition 1: two creditors when default risk is low; one creditor when default risk is high.
- Proposition 2: two creditors when asset complementarity is low; one creditor when complementarity is high.
- Proposition 3: two creditors when outside buyer valuations are low; one creditor when valuations are high.
- The liquidity extension maps low buyer liquidity in bad states to cyclical industries and common-shock industries.

Economic message. Creditor dispersion is optimal when its disciplinary benefit is high relative to its liquidation cost.

low values of β_0 reduce the inefficiency. This liquidation probability, in turn, is low when creditors can require high payments in the good state (because S is low) or when the liquidation value, L_0 , is high (see eq. [7]). While S is lower with the two creditors, L_0 is higher with one creditor. So $\beta_0(2)$ could be greater or less than $\beta_0(1)$.

The following three propositions characterize the conditions under which the two-creditor arrangement is more or less efficient than the one-creditor arrangement.

PROPOSITION 1. The firm borrows from two creditors when default risk is low (θ high) and from one creditor when default risk is high (θ low).

The rationale is simple. When default risk is high, the best way to reduce β_0 is to have a high liquidation value. This is best achieved with one creditor. By contrast, when default risk is low, the more effective way to reduce β_0 is to limit the manager's renegotiation rents, thereby allowing a larger payment in the good state. This is best achieved with two creditors.

The proposition assumes that it is feasible for the firm to borrow from one or from two creditors. However, for low enough θ , neither may be feasible or only one of the alternatives may be feasible. Indeed, there are parameters for which the optimal number of creditors is nonmonotonic in θ . For high values of θ , the firm borrows from two creditors; for intermediate values of θ , the firm borrows from one creditor; and for low values of θ , the firm borrows from two creditors. This last region is possible because at low θ it may not be feasible to borrow from one creditor, but it may be feasible to borrow from two creditors. In particular, this occurs when

$$\begin{aligned} & \theta \frac{y}{2} + (1 - \theta) \left(\frac{\alpha^2 y^2}{4\bar{c}} \right) < K \\ & < \theta \frac{y}{2} + (1 - \theta) \left(\frac{\alpha^2 y^2}{4\bar{c}} \right) + \theta \frac{\Delta}{6} - (1 - \theta) \frac{\alpha^2 \Delta^2}{36\bar{c}}. \end{aligned} \quad (14)$$

PROPOSITION 2. The firm borrows from two creditors when asset complementarity is low (Δ low) and from one creditor when asset complementarity is high (Δ high).

On the one hand, a large Δ is beneficial in that it reduces the manager's renegotiation rent; on the other hand, it is detrimental in that it lowers liquidation payoffs. However, as one can see from (11) and (12), Δ has a first-order effect on renegotiation rents but only a second-order effect on liquidation payoffs. Thus small increases in Δ

Propositions 1–2

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from zero increase efficiency. Beyond a certain point, the liquidation costs outweigh the renegotiation benefits.

PROPOSITION 3. The firm borrows from two creditors when outside buyers place low valuations on the assets (α small) and from one creditor when their valuations are high (α high).

As α increases, liquidation values rise. But, more important, the one-creditor liquidation value rises faster than the two-creditor liquidation value. This effect makes it more attractive for firms to borrow from one creditor when α is high.

One way to interpret the results is to view α as a measure of the ease with which assets can be redeployed. Firms with easily redeployable assets will borrow from one creditor. For example, an office building is likely to be easily redeployable: an outside buyer can get just as much value from the office building as the manager. However, highly redeployable assets probably also do not exhibit much complementarity. Indeed, one of the reasons they can be easily redeployed is that they can be easily combined with other assets. The office building is highly redeployable because it can just as easily be combined with the office equipment of an accounting firm as with the office equipment of a law firm. Thus, while firms with highly redeployable assets might want to borrow from one creditor, those assets may exhibit low asset complementarity, suggesting that the firm should borrow from two creditors. Which effect is more important is an open question.

Another interpretation of α is that it reflects the extent to which outside buyers have liquid resources to purchase the assets. Loosely

Proposition 3 and buyer-liquidity interpretation

5.5 Original equation panels 9–11: security allocation and voting rules

Read:

- Giving one creditor all security reproduces the one-creditor solution.
- Requiring m of n creditors to approve a sale gives $S(m, n) = (1 - m/(n + 1))y$.
- The liquidation payoff is maximized near majority rule, but the optimal voting rule is more stringent than that because stringency lowers renegotiation rents.
- Proposition 4: the optimal voting rule is more stringent for safer firms and for firms whose outside buyers have low valuations.

Economic message. Security interests and voting rules are alternative ways to choose the desired degree of bargaining friction.

OPTIMAL DEBT STRUCTURE

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Equation (18) pins down R_0^b . Combining this with the binding incentive constraint pins down R_0^a :

$$R_0^a + R_0^b = \beta_0[y - S(1)]. \quad (19)$$

Substituting (18) and (19) into (17), we see that β_0 is solved by

$$\beta_0[\theta(y - S(1)) + (1 - \theta)L_0(1)] = K, \quad (20)$$

which is precisely the condition determining $\beta_0(1)$. Thus giving one creditor all of the security replicates the one-creditor solution. It gen-

Security allocation

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First, consider a strategic default. If the voting rule requires m of the n creditors to agree, then the manager will contribute a positive value of y only to those coalitions that have at least m creditors. Since there are $n + 1$ parties in the bargaining, this occurs with probability $1 - [m/(n + 1)]$. Thus

$$S(m, n) = \left(1 - \frac{m}{n + 1}\right)y. \quad (21)$$

Note that the higher m is—the more stringent the voting rule—the lower the manager's surplus from renegotiation.

Similarly, following a liquidity default, the outside buyer will add value only to coalitions with m creditors. So his Shapley value is $\{1 - [m/(n + 1)]\}\alpha y$ and the probability that he bargains is therefore $\{1 - [m/(n + 1)]\}(\alpha y/\bar{c})$. The creditors get the remaining surplus of $[m/(n + 1)]\alpha y$. Thus

$$L_0(m, n) = \frac{m}{n + 1} \left(1 - \frac{m}{n + 1}\right) \frac{\alpha^2 y}{\bar{c}}. \quad (22)$$

In this case, an increase in m has ambiguous effects on L_0 . On the one hand, it increases creditors' payoffs in liquidation because it increases the probability that they will have a positive value to a coalition. On the other hand, it reduces the buyer's Shapley value and discourages him from becoming informed. Differentiating (22) with respect to $m/(n + 1)$, we see that the value of $m/(n + 1)$ that maximizes $L_0(m, n)$ is $1/2$. Thus the maximal value of $L_0(m, n)$ is $\alpha^2 y^2/4\bar{c}$ (which is exactly the payoff in the one-creditor case under the alternative mechanism considered in Sec. III).

A couple of points are worth noting before we continue. First, $m/(n + 1) = 1/2$ is not quite majority rule since $m/n = 1/2(n + 1)/n > 1/2$. However, as $n \rightarrow \infty$, $m/n \rightarrow 1/2$. So we shall consider this majority

Voting-rule payoffs

OPTIMAL DEBT STRUCTURE

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differentiating with respect to v gives the following first-order condition:

$$\theta v^2 \frac{\alpha^2 y}{\bar{c}} - \theta - (1 - \theta)(1 - 2v) \frac{\alpha^2 y}{\bar{c}} = 0. \quad (24)$$

The second-order condition is always satisfied.

First, note that the optimal v , v^* , is greater than $1/2$, the voting rule that maximizes the liquidation value. At $v = 1/2$, (24) is negative, indicating that the efficiency loss could be reduced by increasing v above $1/2$. The reason is that a small increase in v has a second-order effect in lowering the liquidation value but a first-order effect in lowering the renegotiation rent. This in turn lowers β_0 .

Thus, in our model, supermajority voting is always optimal. However, this result is not general; it is driven by the assumption of uniform distribution of c . What is general is the idea that the optimal

Optimal voting rule and Proposition 4

5.6 Original equation panel 12: bankruptcy-law discussion

Read:

- The conclusion emphasizes that debt structure affects negotiations after default.
- The authors note that Chapter 11 imposes voting rules, automatic stays, and exclusivity periods.
- If legal rules make renegotiation too easy, creditors may choose more dispersed debt or collateral structures to restore discipline.
- If bankruptcy law lowers liquidation value or raises the manager's strategic-default payoff, the optimal contract requires a higher liquidation probability.

Economic message. Bankruptcy law changes the background bargaining environment. Contractual debt structure may respond by making renegotiation harder or easier.

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gent voting rules. An optimal debt structure balances two concerns: on the one hand, it should deter defaults; on the other hand, it should not make unavoidable defaults too costly. The model predicts that low-default risk firms, those with strong asset complementarities, and those in noncyclical businesses will tend to borrow from more creditors, spread out security interests, and adopt more stringent voting rules.

One issue that we have not considered is the role of bankruptcy law. We have unrealistically assumed that firms cannot file for bankruptcy and that creditors pursue their liquidation rights as specified in the debt contract. This issue is important in the context of our model because Chapter 11 specifies a detailed set of rules that govern debt renegotiation.

For example, Chapter 11 imposes a particular voting rule on debt renegotiation: one-half of the creditors in number or two-thirds of the creditors in the face value of their claims need to agree. This voting rule could be more or less stringent than the optimal voting rule, v^* . If it is less stringent, creditors might choose other aspects of debt structure that make it harder to renegotiate, for example, by borrowing from more creditors or spreading out security interests. Chapter 11 also includes an automatic stay and an exclusivity period, in which debtors can defer debt payments and remain in Chapter 11 for long periods. This lowers the creditors' liquidation value and increases the payoff from strategic default. In response, the optimal

Conclusion and Chapter 11 discussion

6 Short summary

- Bolton and Scharfstein ask how many creditors a firm should borrow from and how debt governance should be structured.
- The model is an incomplete-contracting model with unverifiable cash flow and contractible physical assets.
- The manager may strategically default by diverting cash.
- Creditors' liquidation rights deter strategic default.
- Liquidation rights are costly when the firm has a true liquidity default.
- The optimal contract chooses the minimum liquidation probability needed to satisfy investor break-even and managerial incentive constraints.
- Debt structure matters because it changes renegotiation payoffs after default.

- One creditor maximizes expected liquidation value because buyers face a single bargaining counterparty.
- Multiple creditors lower the manager's renegotiation rent when assets are complementary.
- Multiple creditors can lower expected liquidation value because outside buyers may not enter bargaining.
- Low-default-risk firms can use multiple creditors because liquidity default is unlikely.
- High-default-risk firms benefit from one creditor because maximizing liquidation value is more important.
- Spreading security interests across creditors reproduces the multiple-creditor solution.
- Giving one creditor all security reproduces the one-creditor solution.
- More stringent voting rules make renegotiation harder and lower the manager's payoff from strategic default.
- The optimal voting rule is more stringent than the rule that maximizes liquidation value.
- Bankruptcy law can alter optimal debt structure by changing liquidation values and strategic-default payoffs.
- The paper's broad message is that debt structure is a governance tool, not merely a financing detail.

7 Conclusion

7.1 Core conclusion

Debt structure is chosen not only to raise funds but also to influence bargaining after default. The number of creditors, the distribution of security interests, and voting rules all shape whether renegotiation is easy or hard. The optimal structure deliberately balances discipline against liquidation efficiency.

7.2 Economic intuition

If renegotiation is too easy, managers can strategically default and keep cash. If renegotiation is too hard, true liquidity defaults destroy too much value. Multiple creditors, dispersed collateral, and supermajority voting can be valuable because they make strategic default costly. But the same devices are harmful when they discourage outside buyers and lower liquidation values.

7.3 Takeaway

The paper turns creditor structure into a central part of corporate finance theory. Bank debt, public debt, security interests, syndication, covenants, and voting rules are not just legal packaging. They determine the ex-post renegotiation game, and that game determines ex-ante incentives.